CS7.603: Topics in Reinforcement Learning (Quiz 2, 30/03/2026)

- Answer all questions
- Duration of 60 minutes (11.45 am to 12.45 pm)
- Total 10 marks (3+4+3)
- Assume any data (if needed) and mention that in the answer
- Use of calculator is allowed

**Q.1)** Find the expression for the optimal control input ($u^*(k) = Fx(k)$), the associated optimal cost ($J^*(x(k))$) and the corresponding state trajectory for the dynamical system given below using
i) Policy Iteration (PI)
ii) Value iteration (VI)

$$x(k+1) = 4x(k) + 0.1u(k) \tag{1}$$

The cost to be minimized ($J(x(k))$) is given below,

$$J(x(0)) = \frac{1}{2}\sum_{k=0}^{k=\infty}(Qx^2(k) + Ru^2(k)) \tag{2}$$

$x(0) = 5$, $Q = 1$, $R = 0.1$.

Please provide the results of the first 3 iterations for PI and VI. Assume suitable initial conditions for the required variables.
(2+1=3 Marks)

**Q.2)** State and explain Hamilton-Jacobi-Bellman (HJB) equation (derivation not needed). Find the optimal control input for the system given below using HJB equation.

$$\frac{dx(t)}{dt} = 2x(t) + 0.5u(t) \tag{3}$$

The cost to be minimized is given below with $t_f = 0.2$ and $x(0) = 20$.

$$J(x(0)) = x(t_f)^2 + \int_0^{t_f} 10x(t)^2 + 2u(t)^2\, dt \tag{4}$$

(1+3 = 4 Marks)

**Q.3)** For the linear dynamical system given below,

$$x(k+1) = 0.5x(k) + u(k) \tag{5}$$

the $Q$ value function is as given below (here $\alpha$, $\beta$ and $\lambda$ are constants greater than zero)

$$Q(x(k), u(k)) = \alpha x^2(k) + \beta u^2(k) + \lambda x(k)u(k) \tag{6}$$

Find the optimal control input ($u^*(k)$) and the corresponding state trajectory ($x^*(k)$) for $k = 1, 2, 3$ with $x(0) = 4$ ?
(3 Marks)



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